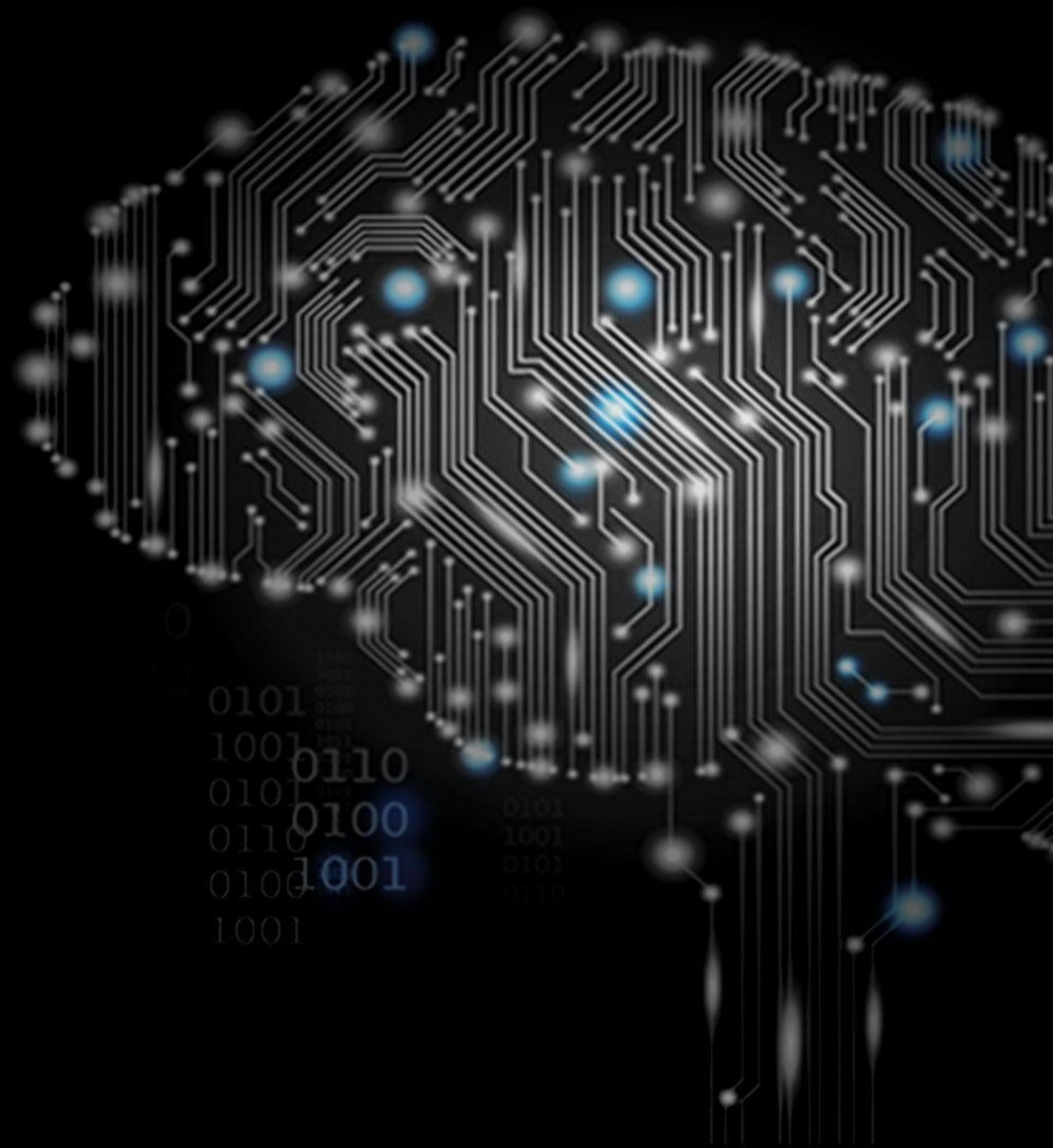
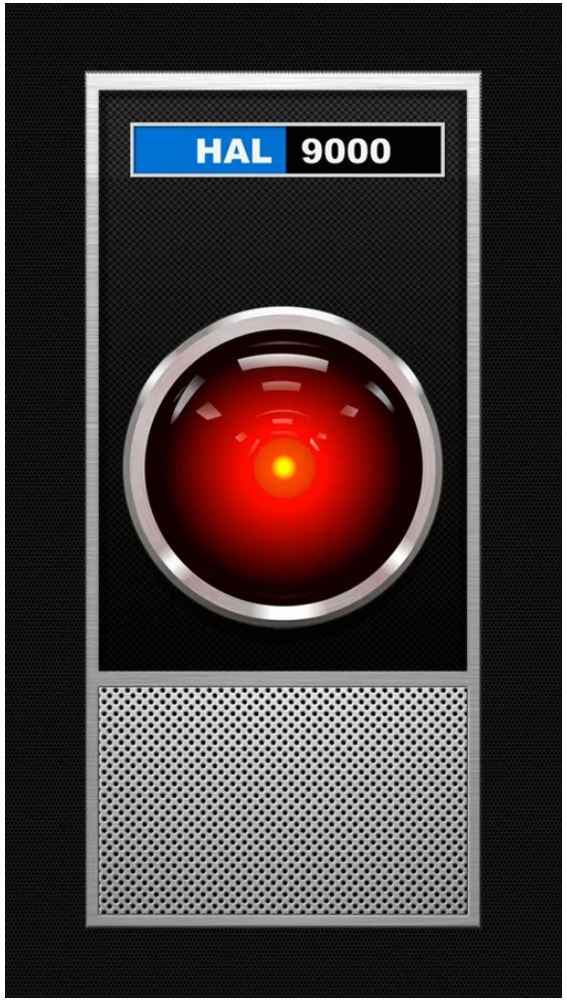
# Introduction to Generative AI

## Course Introduction



# Course Introduction

- The field of generative AI (GAI) and other AI-related technologies are in a constant state of churn
  - New developments, techniques, and technologies are rapidly being developed and deployed
  - This has resulted in a lot of hype and misinformation from a variety of sources
- This course is designed to provide a solid technical overview of GAI as of 2026
  - The course is not a technical deep dive, but a survey of the major topics that are current in the field of GAI
  - The objective of this course is to introduce the main GAI technologies that software developers and engineers need to be familiar with

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# Course Topics

- The AI Landscape
- Transformers and Large Language Models
- Diffusion Models
- Prompt Engineering
- Retrieval-Augmented Generation (RAG)
- Fine-Tuning and Model Adaptation
- AI Agents and Tool-Using Models
- Evaluation, Safety and Governance

Image Credit: https://medium.com/@ai.agenda

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# Logistics

- Repository for class materials
  - https://github.com/ExgnosisClasses/2609-GenAI-Feb-24
- Hours 9am – 5pm CST, Feb 24-35
  - Lunch is 12-1 CST
  - Two 15 breaks, one mid-morning and one mid-afternoon
- Labs repository
  - https://github.com/ExgnosisClasses/2609-GenAi-Labs
- Lab machines
  - VMs will be provided for the labs
  - Lab #1 will walk you through how to log on to your VM

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# Instructor – Rod Davison

- 50+ years experience in IT
- 40+ years experience in AI
  - Designed and developed AI prototypes
    - NLU, expert systems, machine translation, agents and more
  - Research director for AI companies
- Academic background
  - Theoretical math, psychology, anthropology, cognitive science
- Consulting
  - Finance, Defence, Insurance, Government, Fortune 100
  - Multiple areas

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# Introductions

- In order for me to ensure the class meets your needs
- Please introduce yourself and tell me:
  - Your professional background and skill set (developer, BA, etc.)
  - Your experience in using AI and GAI
  - Any specific goals or objectives for the class

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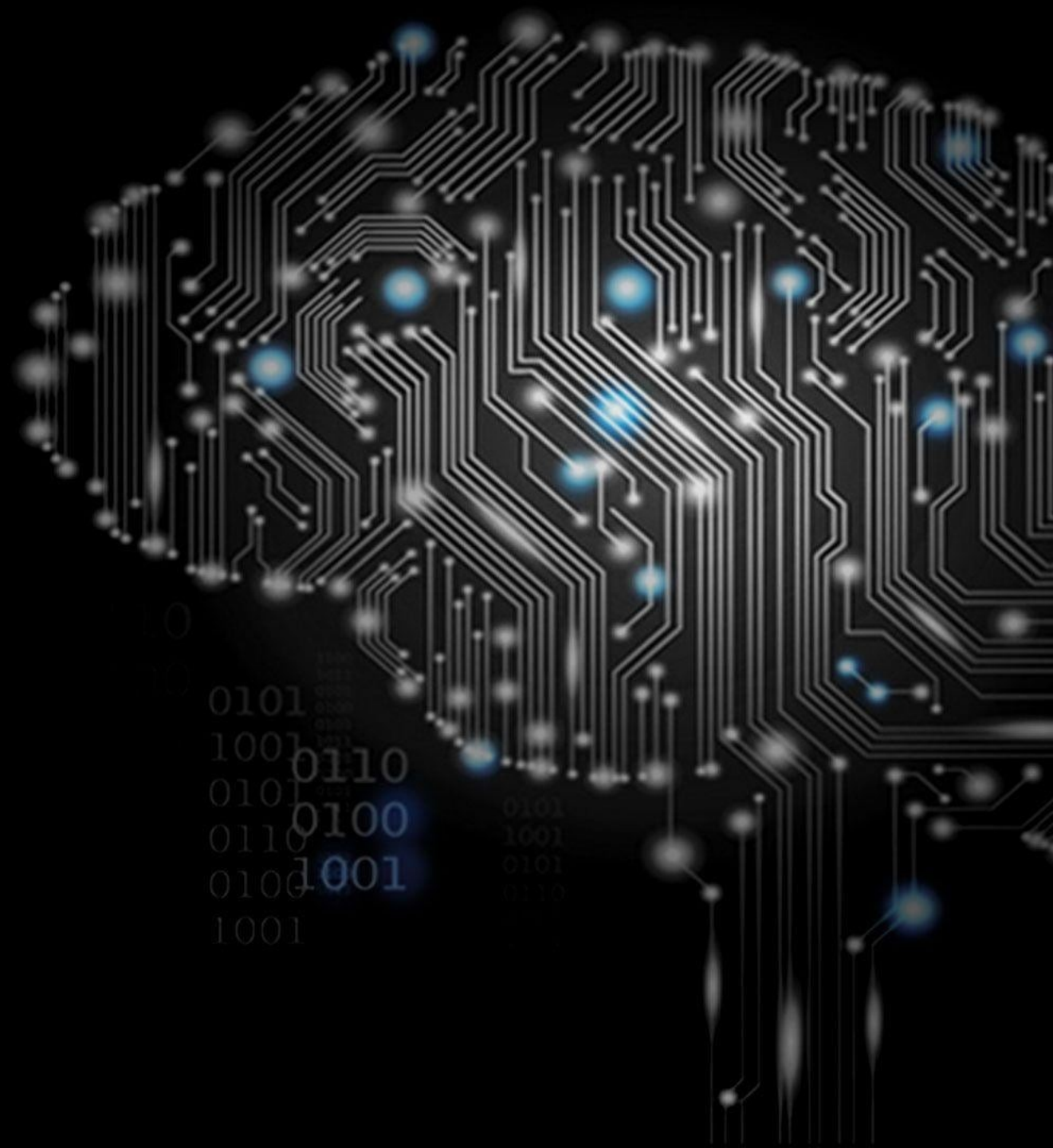

**Questions?**